# ECE 3355: Electronics

Section 10819
Spring 2019

## Exam 2

Version B - Spring 2019

**Aril 16, 2019**

Do *not* open the exam until instructed to do so. Answer the questions in the spaces provided on the question sheets. If you run out of room for an answer, continue on the back of the page. This is a closed-book/notes exam and you *may* use a calculator. **You will have 1 hr. 15 min. to finish the exam.**

Student's Name: _____

| Question | Points | Score |
|---|---|---|
| 1 | 25 | |
| 2 | 25 | |
| 3 | 50 | |
| Total: | 100 | |

From Table 6.2:

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$$

$$i_E = \frac{i_C}{\alpha} = \left(\frac{I_S}{\alpha}\right) e^{v_{BE}/V_T}$$

Note: For the pnp transistor, replace $v_{BE}$ with $v_{EB}$.

$$i_C = \alpha i_E$$

$$i_C = \beta i_B$$

$$\beta = \frac{\alpha}{1-\alpha}$$

$$i_B = (1-\alpha)\, i_E = \frac{i_E}{\beta+1}$$

$$i_E = (\beta+1)\, i_B$$

$$\alpha = \frac{\beta}{\beta+1}$$

$V_T =$ thermal voltage $= \dfrac{kT}{q} \simeq 25\,\text{mV}$ at room temperature

Summary of Table 6.4 (Small Signal Model Parameters)

Model Parameters in Terms of DC Bias Currents

$$g_m = \frac{I_C}{V_T} \quad r_e = \frac{V_T}{I_E} = \alpha\frac{V_T}{I_C} \quad r_\pi = \frac{V_T}{I_B} = \beta\frac{V_T}{I_C} \quad r_o = \frac{|V_A|}{I_C}$$

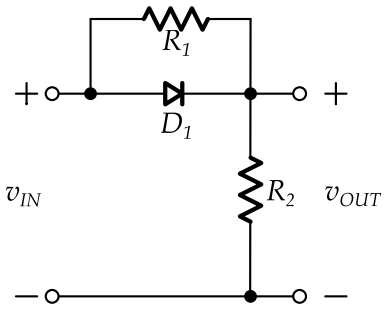
In Terms of $g_m$

$$r_e = \frac{\alpha}{g_m} \qquad r_\pi = \frac{\beta}{g_m}$$

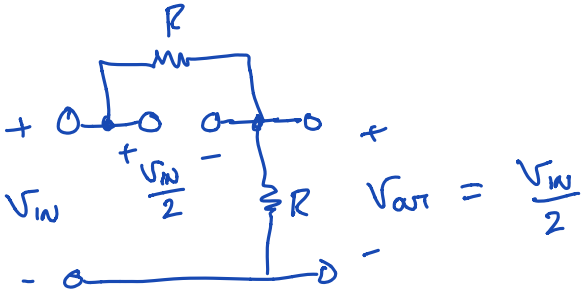
In Terms of $r_e$

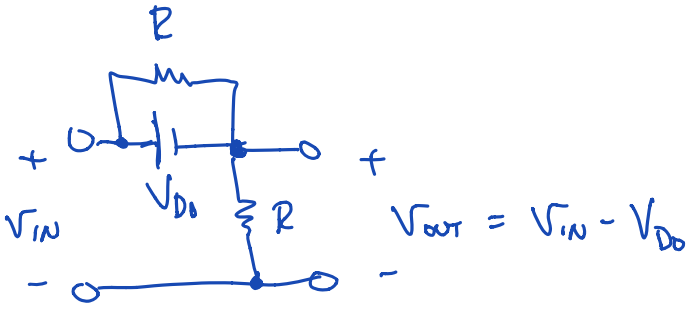
$$g_m = \frac{\alpha}{r_e} \quad r_\pi = (\beta+1)\, r_e \qquad g_m + \frac{1}{r_\pi} = \frac{1}{r_e}$$

1. For the circuit below, assume $V_D = 0.7$ volts and $r_D = 0$.




(a) (10 points) Write an expression for $v_{OUT}$ in terms of $v_{IN}$. w| $R_1 = R_2$



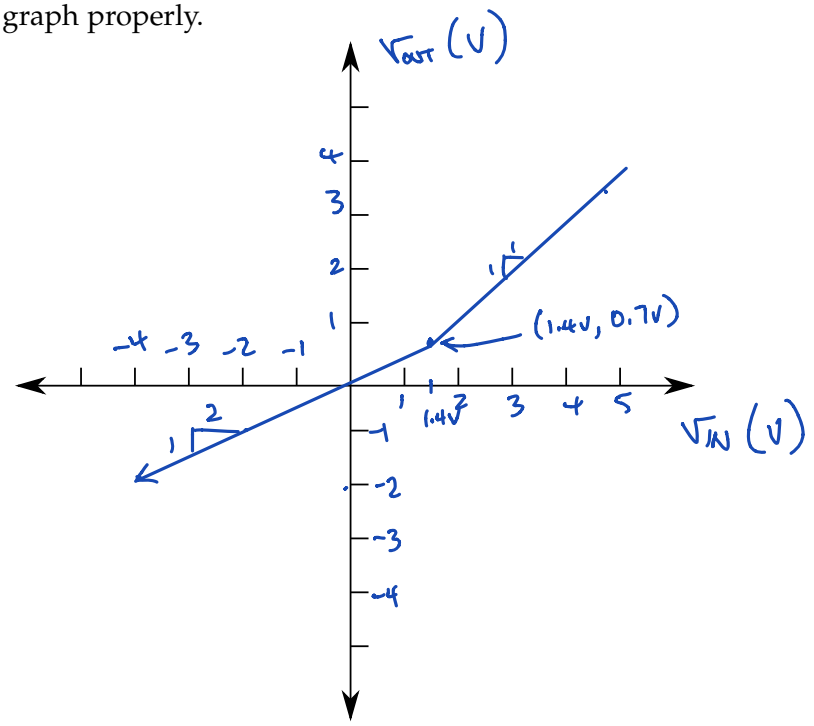




Diode is forward biased when $\frac{V_{IN}}{2} \geq V_{D0}$ or $V_{IN} \geq 2V_{D0}$

so

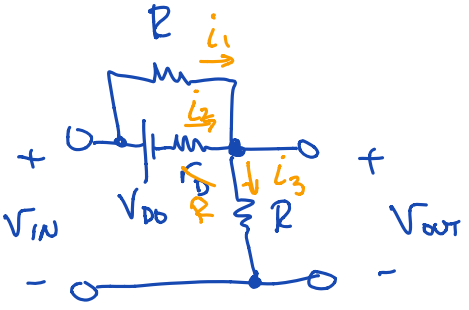
$$V_{OUT} = \begin{cases} \frac{V_{IN}}{2} & V_{IN} < 2V_{D0} \\ V_{IN} - V_{D0} & V_{IN} \geq 2V_{D0} \end{cases}$$

(b) (5 points) Sketch $v_{OUT}$ and $v_{IN}$ for $-4$ volts $< v_{IN} < 4$ volts on the graph provided. Be sure to label the graph properly.




(c) (10 points) Now, assume that $r_D$ is no longer zero, $R = r_D$, and $V_{D0} = 0.5$ volts. Write an expression for $v_{OUT}$ in terms of $v_{IN}$.

when diode is forward biased:




$$i_1 = \frac{V_{IN} - V_{OUT}}{R}$$

$$i_2 = \frac{V_{IN} - V_{D0} - V_{OUT}}{R}$$

$$i_3 = \frac{V_{OUT}}{R}$$

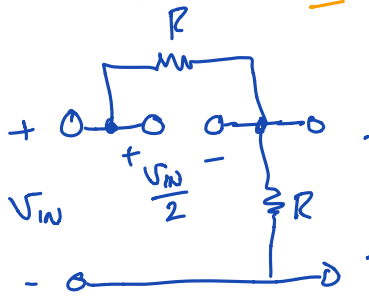
$$i_3 = i_1 + i_2$$

$$\frac{V_{OUT}}{R} = \frac{V_{IN} - V_{OUT}}{R} + \frac{V_{IN} - V_{D0} - V_{OUT}}{R}$$

$$3V_{OUT} = 2V_{IN} - V_{D0}$$

$$V_{OUT} = \frac{2V_{IN} - V_{D0}}{3}$$

when diode is not forward biased

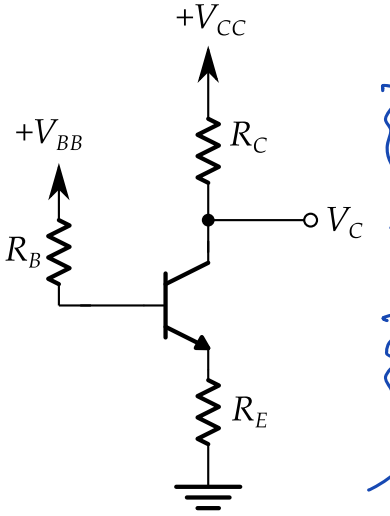



$$V_{OUT} = \frac{V_{IN}}{2}$$

So:

$$V_{OUT} = \begin{cases} \frac{V_{IN}}{2} & V_{IN} < 2V_{D0} \\ \frac{2V_{IN} - V_{D0}}{3} & V_{IN} \geq 2V_{D0} \end{cases}$$

2. Use the following circuit to complete this problem.




b/c $R_C = R_E$ and $I_E \cong I_C$, these two voltages will be about the same and $V_{CE} < 0.2V$ → Hence we will always be in saturation

(a) (10 points) If $R_E = R_C = 1\,\text{k}\Omega$, $V_{BB} = 9$ volts, $\beta = 100$, and $V_{CC} = 16$ volts, is it possible to choose a value for $R_B$ so that the transistor is active and $V_C = V_{CC}/2$? Justify your answer to receive credit for this problem.

when $V_c = \frac{V_{cc}}{2}$, $I_c R_c = \frac{V_{cc}}{2}$

Also,

$I_E = \frac{I_c}{\alpha}$ and $V_E = I_E R_E = \frac{I_c}{\alpha} R_E$

Since $R_E = R_C$, $V_E = \frac{I_c}{\alpha} R_E > I_c R_c = \frac{V_{cc}}{2} = V_c$

That means that $V_{CE} = V_c - V_E$ is negative and transistor will always be in saturation.

(b) (15 points) Design: Find values for $R_E$ and $R_C$ so that $V_C = V_{CC}/2$ and $r_\pi = 312.5\,\Omega$ when $R_B = 90\,\text{k}\Omega$, $V_{BB} = 9$ volts, $\beta = 100$, and $V_{CC} = 16$ volts.

$$r_\pi = \frac{V_T}{I_B} \Rightarrow I_B = \frac{V_T}{r_\pi} = \frac{25\text{mV}}{312.5\Omega} = 0.08\text{mA}$$

Hence $I_C = \beta I_B = 8\text{mA}$

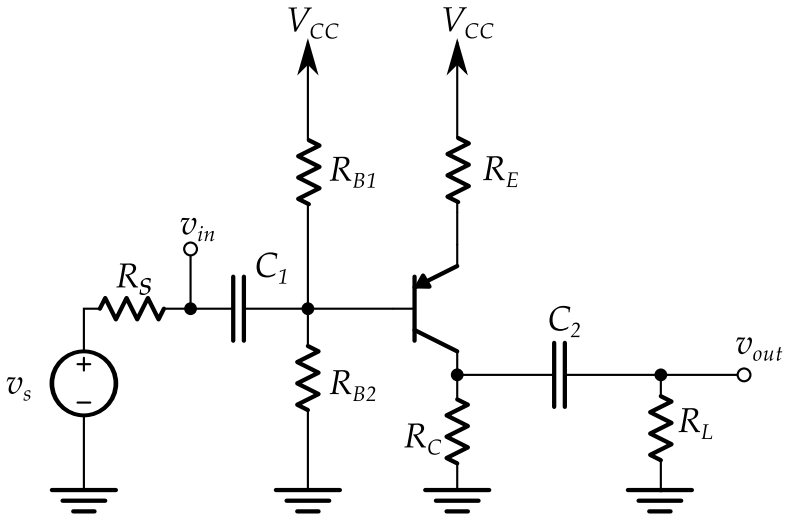
$$V_{CC} - I_C R_C = \frac{V_{CC}}{2} \Rightarrow R_C = \frac{V_{CC}}{2I_C} = \frac{16\text{V}}{2 \cdot 8\text{mA}} = 1\text{K}\Omega$$

$$V_{BB} = I_E R_E + V_{BE} + I_B R_B$$

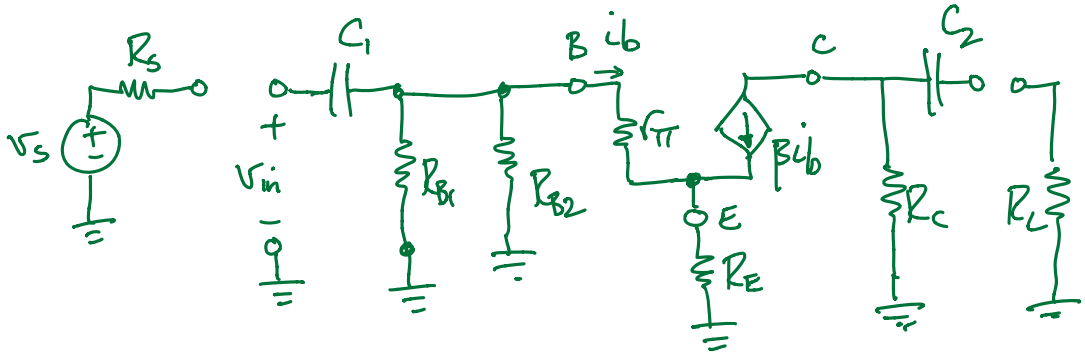
$$R_E = \frac{V_{BB} - V_{BE} - I_B R_B}{I_E} = \frac{9\text{V} - 0.7\text{V} - (0.08\text{mA})(90\text{K}\Omega)}{101 \cdot 0.08\text{mA}}$$

$$= 136\,\Omega$$

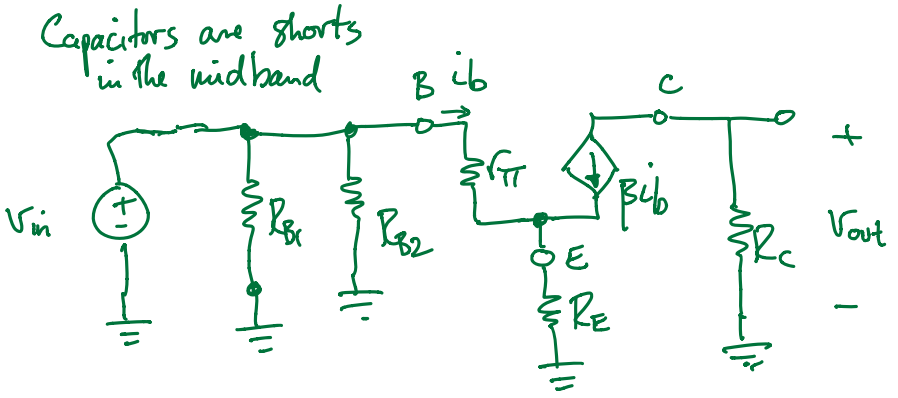
3. Use the following circuit to complete this problem.




(a) (10 points) Assuming that the transistor is in the active mode, sketch the small signal model for the circuit.




(b) (10 points) Write an expression for the *midband gain* of the circuit, $\frac{v_{out}}{v_{in}}$, with the load and source removed. The solution should be expressed in terms of resistors, $\beta$, and $r_\pi$.




$$v_{out} = -\beta i_b R_C$$

$$v_{in} = i_b r_\pi + (\beta+1) i_b R_E$$

$$\boxed{\frac{v_{out}}{v_{in}} = \frac{-\beta R_C}{r_\pi + (\beta+1) R_E}}$$

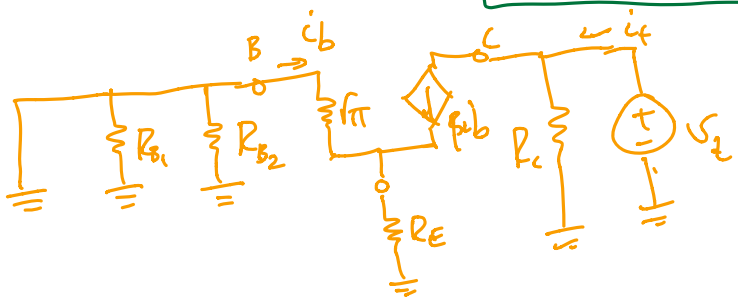
(c) (20 points) Find expression for the input and output resistance for the *midband* with the source and load removed.

$$R_{in} = \frac{v_{in}}{i_{in}} = \frac{v_{in}}{\frac{v_{in}}{R_{B1}} + \frac{v_{in}}{R_{B2}} + i_b} = \frac{v_{in}}{\frac{v_{in}}{R_{B1}} + \frac{v_{in}}{R_{B2}} + \frac{v_{in} i_b}{v_{in}}}$$

$$= \frac{1}{\frac{1}{R_{B1}} + \frac{1}{R_{B2}} + \frac{1}{\frac{v_{in}}{i_b}}} = R_{B1} \| R_{B2} \| \left(\frac{v_{in}}{i_b}\right)$$

you could also have recognized this from the circuit

$$\boxed{R_{in} = R_{B1} \| R_{B2} \| (r_\pi + (\beta+1) R_E)}$$




b/c $v_{in} = 0$, $i_b r_\pi + (\beta+1) i_b R_E = 0$ and $i_b = 0$

Hence, $v_t = i_t \cdot R_C$ and $\boxed{R_{out} = \frac{v_t}{i_t} = R_C}$

(d) (5 points) Write expressions for the resistances seen by the capacitors $C_1$ and $C_2$ in terms of $R_s$, $R_{in}$, $R_{out}$, and $R_L$.

Now that we have found $R_{in}$, $R_{out}$, and $A_v$, we can use the simplified model:




so, $\boxed{R_{C_1} = R_s + R_{in}}$ and $\boxed{R_{C_2} = R_L + R_{out}}$

(e) (5 points) Make a rough sketch of a possible Bode magnitude plot for this circuit (please label slopes).

